

BRACT's

Vishwakarma Institute of Information Technology, Pune Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 3 Practical No: 3 Roll no: 22 PRN No: 12414149

Subject Name:

Name of Student: Prathistha Yadav

Assignment-1

Problem Statement:

Shell Programming: Implement student database using shell and awk programming to perform following operations:

1. Create database
2. View database
3. Insert a record
4. Delete a record
5. Modify a record
6. Display result of a particular student
7. Exit

Algorithm:

Step 1: Start the program.

Step 2: Create a database file named student.txt.

Step 3: Display the following menu:

Create Database

View Database

Insert a Record

Delete a Record

Modify a Record

Display Result of a Particular Student

Exit

Step 4: Read the user's choice.

Step 5: Perform the selected operation:

Create Database: Create student.txt and add the field headings.

View Database: Use AWK to read and display all student records.

Insert Record: Accept roll number, name and marks from the user and append the record to the database.

Delete Record: Accept the roll number and use AWK to remove the matching record.

Modify Record: Accept the roll number and new student details, then use AWK to replace the old record.

Display Result: Search for the given roll number using AWK, calculate total and percentage, and display PASS/FAIL.

Exit: Terminate the program.

Step 6: After completing the selected operation, display the menu again.

Step 7: Continue until the user selects **Exit**.

Step 8: Stop the program.

Code-

```
#!/bin/bash

DB="student.txt"

while true
do
echo "=====
echo " STUDENT DATABASE"
echo "=====
echo "1. Create Database"
echo "2. View Database"
echo "3. Insert a Record"
echo "4. Delete a Record"
echo "5. Modify a Record"
echo "6. Display Result"
echo "7. Exit"
echo "=====

read -p "Enter your choice: " choice
```

```
case $choice in
```

```
1)
```

```
echo "RollNo|Name|Marks1|Marks2|Marks3" > $DB
```

```
echo "Database created successfully."
```

```
::
```

```
2)
```

```
echo "----- DATABASE -----"
```

```
cat $DB
```

```
::
```

```
3)
```

```
read -p "Enter Roll Number: " roll
```

```
read -p "Enter Name: " name
```

```
read -p "Enter Marks 1: " m1
```

```
read -p "Enter Marks 2: " m2
```

```
read -p "Enter Marks 3: " m3
```

```
echo "$roll|$name|$m1|$m2|$m3" >> $DB
```

```
echo "Record inserted successfully."
```

```
::
```

```
4)
```

```
read -p "Enter Roll Number to delete: " roll
```

```
awk -F'|' -v r="$roll" 'NR==1 || $1!=r' $DB > temp.txt
```

```
mv temp.txt $DB
```

```
echo "Record deleted successfully."
```

```
::
```

```
5)
```

```
read -p "Enter Roll Number to modify: " roll
```

```
read -p "Enter New Name: " name
```

```
read -p "Enter New Marks 1: " m1
```

```
read -p "Enter New Marks 2: " m2
```

```
read -p "Enter New Marks 3: " m3
```

```
awk -F'|' -v r="$roll" -v n="$name" -v a="$m1" -v b="$m2" -v c="$m3" '
```

```
BEGIN { OFS="|" }
NR==1 { print; next }
$1==r {
$2=n
$3=a
$4=b
$5=c
}
{ print }
' $DB > temp.txt
```

```
mv temp.txt $DB
echo "Record modified successfully."
;;
```

```
6)
read -p "Enter Roll Number: " roll
```

```
awk -F'|' -v r="$roll" '
NR>1 && $1==r {
total=$3+$4+$5
percentage=total/3
```

```
print "====="
print " STUDENT RESULT"
print "====="
print "Roll Number :", $1
print "Name :", $2
print "Marks 1 :", $3
print "Marks 2 :", $4
print "Marks 3 :", $5
print "Total :", total
printf "Percentage : %.2f%%\n", percentage
```

```
if (percentage >= 40)
print "Result : PASS"
else
print "Result : FAIL"
```

```
found=1
}
```

```

END {
if(!found)
print "Student not found."
}
' $DB
;;

7)
echo "Program terminated."
exit
;;

*)
echo "Invalid choice!"
;;

esac

echo ""
done

```

OUTPUT:

```

root@DESKTOP-GEURPBV: /r
6. Display Result of a Student
7. Exit
=====
Enter your choice: 2
===== STUDENT DATABASE =====
RollNo   Name           Marks1   Marks2   Marks3
===== STUDENT DATABASE =====
1. Create Database
2. View Database
3. Insert a Record
4. Delete a Record
5. Modify a Record
6. Display Result of a Student
7. Exit
=====
Enter your choice: 3
Enter Roll Number: 22
Enter Name: prathistha yadav
Enter Marks in Subject 1: 89
Enter Marks in Subject 2: 92
Enter Marks in Subject 3: 94
Record inserted successfully.
===== STUDENT DATABASE =====
1. Create Database
2. View Database
3. Insert a Record
4. Delete a Record
5. Modify a Record
6. Display Result of a Student
7. Exit
=====
Enter your choice: 6
Enter Roll Number: 22
===== STUDENT RESULT =====

```

```
root@DESKTOP-GEURPBV: /n x + v
Enter your choice: 6
Enter Roll Number: 22
```

```
===== STUDENT RESULT =====
Roll Number : 22
Name       : prathistha yadav
Subject 1  : 89
Subject 2  : 92
Subject 3  : 94
Total      : 275
Percentage : 91.67%
Result     : PASS
```

```
===== STUDENT DATABASE =====
1. Create Database
2. View Database
3. Insert a Record
4. Delete a Record
5. Modify a Record
6. Display Result of a Student
7. Exit
```

```
=====
Enter your choice: 3
Enter Roll Number: 23
Enter Name: Miranjan K
Enter Marks in Subject 1: 90
Enter Marks in Subject 2: 92
Enter Marks in Subject 3: 98
Record inserted successfully.
```

```
===== STUDENT DATABASE =====
1. Create Database
2. View Database
3. Insert a Record
4. Delete a Record
5. Modify a Record
6. Display Result of a Student
```

```
root@DESKTOP-GEURPBV: /mnt/c/Users/Prathistha Yadav# touch exe.sh
root@DESKTOP-GEURPBV: /mnt/c/Users/Prathistha Yadav# nano exe.sh
root@DESKTOP-GEURPBV: /mnt/c/Users/Prathistha Yadav# chmod +x exe.sh
Command 'chmod' not found, did you mean:
  command 'chmod' from deb coreutils-from-gnu (0.0.0-ubuntu25)
  command 'chmod' from deb coreutils-from-utils (0.0.0-ubuntu25)
Try: apt install <deb name>
root@DESKTOP-GEURPBV: /mnt/c/Users/Prathistha Yadav# chmod +x exe.sh
root@DESKTOP-GEURPBV: /mnt/c/Users/Prathistha Yadav# ./exe.sh
```

```
===== STUDENT DATABASE =====
1. Create Database
2. View Database
3. Insert a Record
4. Delete a Record
5. Modify a Record
6. Display Result of a Student
7. Exit
```

```
=====
Enter your choice: 1
Creating Database...
Database created successfully.
```

```
===== STUDENT DATABASE =====
1. Create Database
2. View Database
3. Insert a Record
4. Delete a Record
5. Modify a Record
6. Display Result of a Student
7. Exit
```

```
=====
Enter your choice: 2
===== STUDENT DATABASE =====
RollNo  Name      Marks1  Marks2  Marks3
===== STUDENT DATABASE =====
```

```
root@DESKTOP-GEURPBV:/n  X + | v

===== STUDENT RESULT =====
Roll Number : 23
Name       : Miranjan K
Subject 1  : 94
Subject 2  : 92
Subject 3  : 98
Total     : 284
Percentage : 94.67%
Result    : PASS

===== STUDENT DATABASE =====
1. Create Database
2. View Database
3. Insert a Record
4. Delete a Record
5. Modify a Record
6. Display Result of a Student
7. Exit
=====
Enter your choice: 2

===== STUDENT DATABASE =====
RollNo  Name      Marks1  Marks2  Marks3
22      prathistha yadav  89      92      94
23      Miranjan K      94      92      98
25      samiksha P      89      92      91

===== STUDENT DATABASE =====
1. Create Database
2. View Database
3. Insert a Record
4. Delete a Record
5. Modify a Record
6. Display Result of a Student
7. Exit
=====
Enter your choice: |
```